

Unit 5, Lesson 6: Using Two- and Three-Column Tables of Equivalent Ratios



Benchmarks

MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.



Learning Targets

- ☐ I can use two- and three-column tables to display equivalent ratios.
- ☐ I can solve problems using two- and three-column tables of equivalent ratios.



5.6.1 Warm-Up: Different Ways of Expressing Division

In many Latin American countries and other countries around the world, the colon (:) is used instead of $\div$ to express division. Therefore, the expression “28 divided by 4” would be expressed as 28:4 instead of $28 \div 4$.

1. Using what you have learned about ratios, explain how 28:4 and $28 \div 4$ can be used to express the same value.

2. Write one question that comes to your mind after reading the information shown.

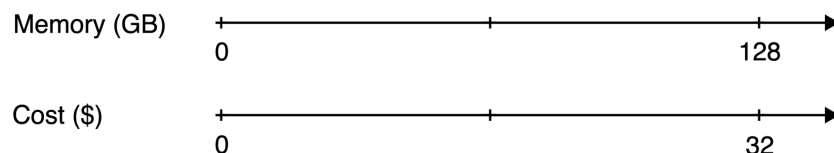


5.6.2 Exploration: Double Number Lines Versus Two-Column Tables

Work with your partner on the following.

In 2016, 128 gigabytes (GB) of portable computer memory cost \$32.

A double number line diagram that represents the situation is shown.



1. One set of tick marks has already been drawn to show the result of multiplying 128 and 32 each by $\frac{1}{2}$. Label the amount of memory and the cost for these tick marks.
2. Next, keep multiplying by $\frac{1}{2}$ and drawing and labeling new tick marks, until you can no longer clearly label each new tick mark with a number.
3. Discuss with your partner whether or not it's possible to clearly represent the cost of 1 GB of memory on the double number line diagram.
4. If it is possible, circle the cost of 1 GB of memory on the double line diagram. If it is not possible, explain why not.

5. A table representing the same situation is shown. Work with your partner to find the cost of 1 GB. Use as many or few rows as needed.

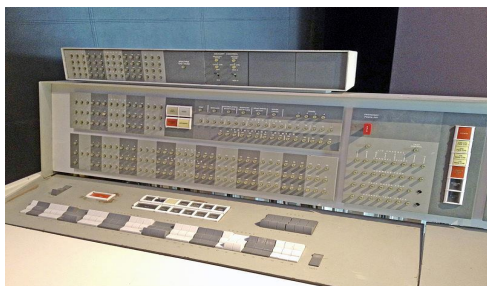
Memory (gigabytes)	Cost (dollars)
128	32

6. Explain your steps for determining the cost per gigabyte using the table.
7. Use either the double line diagram or the ratio table to determine the cost of 512 GB of portable computer memory at this rate.
8. Discuss with your partner which representation of the equivalent ratios you preferred using to find the costs of 1 GB and 512 GB. Summarize your discussion.



The IBM 7094 was regarded as one of the most powerful advanced mainframe computers in the early 1960s. NASA and the Air Force used it in the Gemini and Apollo space programs.

Despite its large size and historical impact, the IBM 7094 only had a core memory of about 150 kilobytes (KB). That's only enough to manage a few Microsoft Word documents.



Today, you can get a small USB flash drive like the one shown for less than \$20 that can store 4 gigabytes (GB) of memory.



That's almost 27,000 times the memory as the IBM 7094, and you can fit it in your pocket!



5.6.3 Guided Instruction: Using Three-Column Ratio Tables

Ratios can be used to make part-to-part or part-to-whole comparisons. When comparing parts of a mixture to the whole, three-column ratio tables can be useful to represent the relationships between quantities.

1. Kiara is practicing her free-throw shot at basketball practice. She averages 4 made shots for every 2 missed shots. The table shown can be used to represent this situation.

Number of Shots Made	Number of Shots Missed	Total Shots Taken
4	2	

- a. When determining this rate during practice, Kiara missed 8 free-throw shots. Complete the second row of the table to show how many shots she made and how many total shots she took.
- b. How was each value in the second row determined?

2. Kiara had the opportunity to take 3 free-throw shots in her most recent game, and her accuracy remained the same as in practice. Complete the third row of the table to show how many free-throw shots she made and missed in the game.

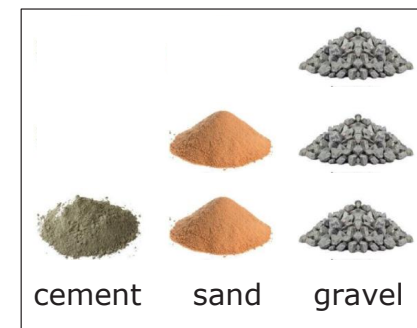
a. How was each value in the third row determined?

b. Complete the statement.

For every 3 free-throw shots taken, Kiara makes _____ shot(s) and misses _____ shot(s) on average. The ratio of missed shots to made shots to total shots is _____ : _____ : _____.

3. Three-column tables can also be useful when comparing mixtures that contain more than two parts.

Concrete can be created by mixing 1 part cement with 2 parts sand and 3 parts gravel. These dry ingredients are first combined before being mixed with water.



- a. Complete the table to determine equivalent mixtures of concrete. Show how the values in each row are determined using arrows and by indicating the factor used.

Cement	Sand	Gravel
1	2	3
		1
6.5		
	10	

- b. Marc uses one-gallon buckets to measure out the quantities needed in his concrete mixture. How many gallons of dry mix will he have in one batch of concrete?



5.6.4 Guided Practice: Using Two- and Three-Column Ratio Tables

1. A particular shade of maroon paint calls for 5 cups of red paint to be mixed with 3 cups of blue paint. Complete the table to find the quantities needed for different batches of paint.

Red Paint (cups)	Blue Paint (cups)	Maroon Paint (cups)
5	3	
	12	
		2
2.5		
		80

- a. If the mixture in the first row represents one batch of maroon paint, how many batches are in 80 cups of maroon paint?
- b. Explain how this relates to the factor used to complete the last row in the table.

2. There are 16 cups in one gallon. Use this information and the table below to determine how many gallons of paint the 80-cup mixture of maroon paint makes.

Cups	Gallons

3. In April, Kyla went for a walk on 4 days for every 1 day that she did not walk. There are 30 days in April. Use this information to complete the ratio table.

Walking Days	4	
Non-Walking Days	1	
Total Days		30

Complete the statement.

Throughout the year, Kyla continued walking 4 days for every 1 day that she did not walk. Based on this, in November, Kyla walked_____ of the 30 days.



5.6.5 Wrap-Up: Ticket Out the Door

1. Julio participated in a walk-a-thon to raise money for cancer research. Walking at a constant speed, Julio traveled 6.4 miles in the first 2 hours of the walk-a-thon. His goal is to walk for 7 hours. Use the table to find how far Julio will have walked if he reaches his 7-hour goal. Show how the values in each row are determined.

Hours	Miles Walked
2	6.4

Unit 5, Lesson 7: Digging Deeper Into Unit



Benchmarks

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.



Learning Targets

- ☐ I can determine a unit rate when given different ratios.
- ☐ I can interpret unit rates in real-world contexts.
- ☐ I can use unit rates to solve problems.